

LAPORAN PRAKTIKUM DML

Diajukan untuk memenuhi salah satu tugas praktikum Mata kuliah
Basis Data Praktek

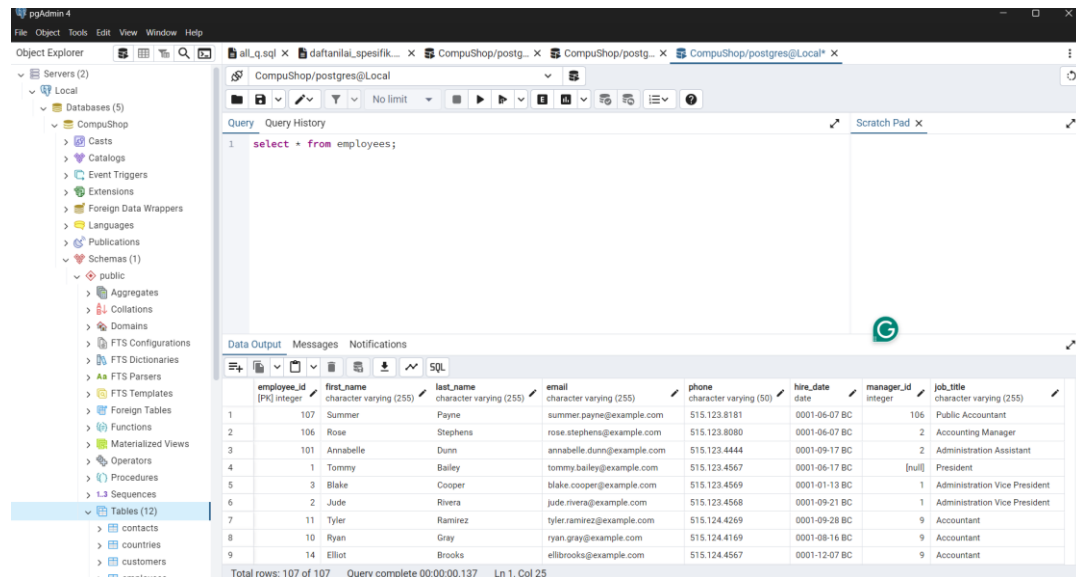


Disusun Oleh:
Muhammad Raihan Pratama (231511055)
Jurusan Teknik Komputer dan Informatika

Program Studi D-3 Teknik Informatika
Politeknik Negeri Bandung
2024

1. Select all columns

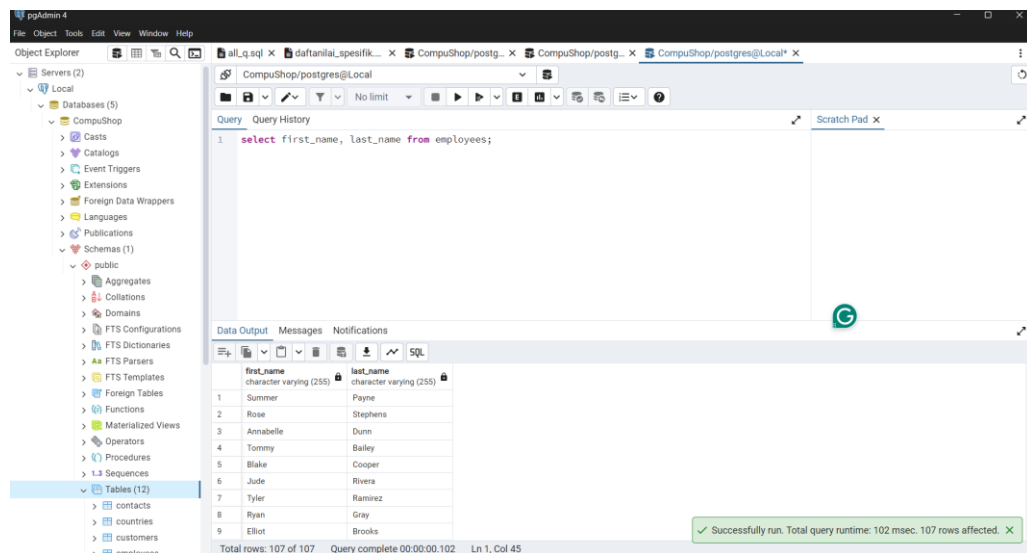
Perintah untuk menampilkan seluruh data yang ada pada tabel



2. Select Columns

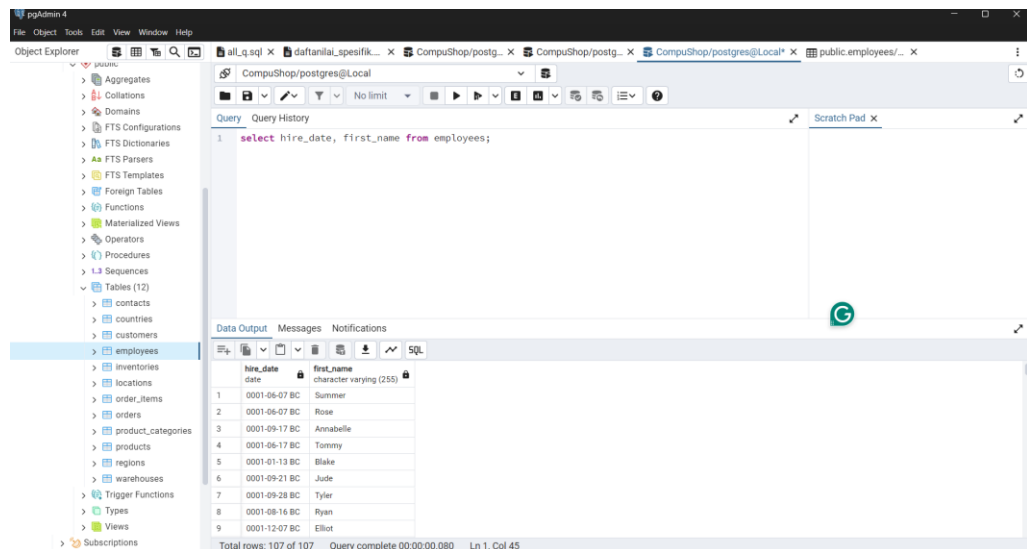
Menampilkan sebagian data berdasarkan nama kolom pada tabel

- First_name and Last_name



- Salary, hire_date, first_name

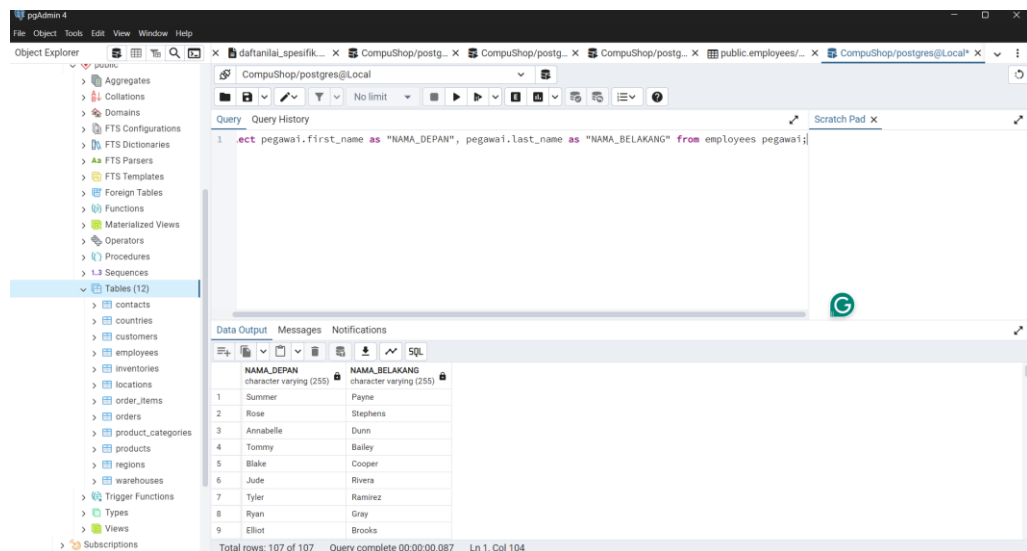
Catatan: tidak terdapat kolom salary pada tabel



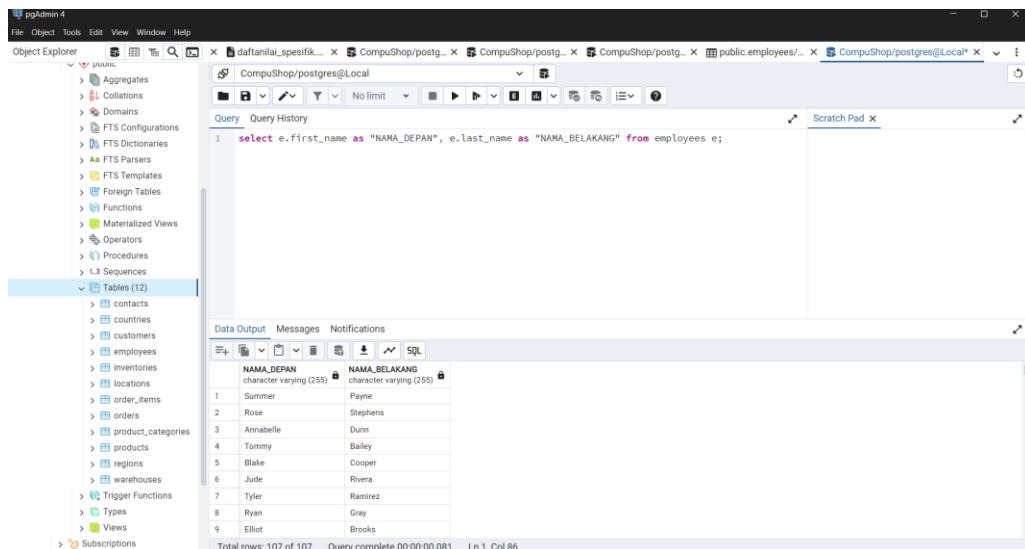
3. Table alias

Memanggil nama tabel dengan sebutan lain

- employee as pegawai

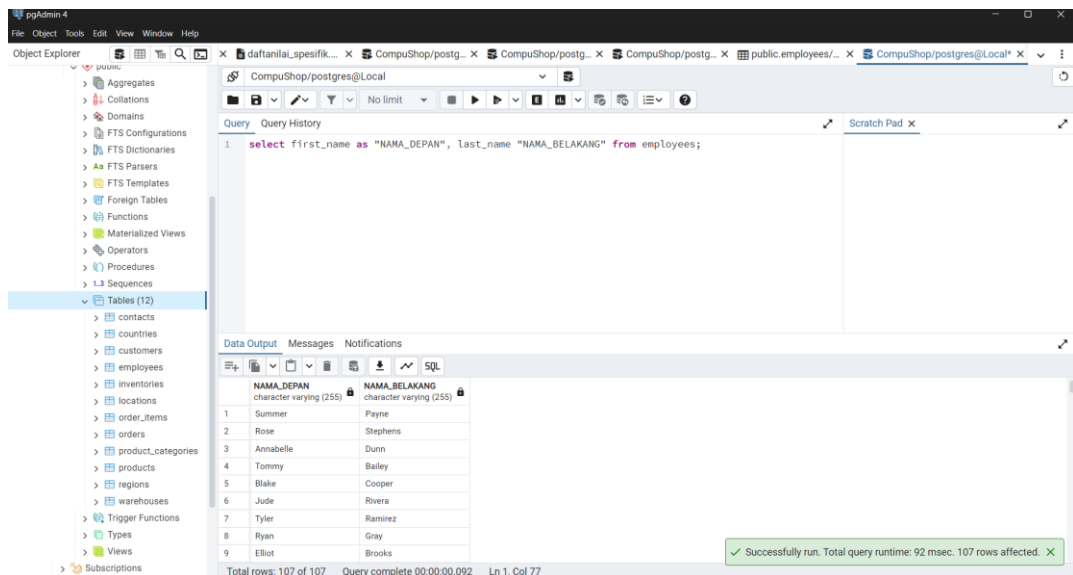


- employee as e



4. Column Alias

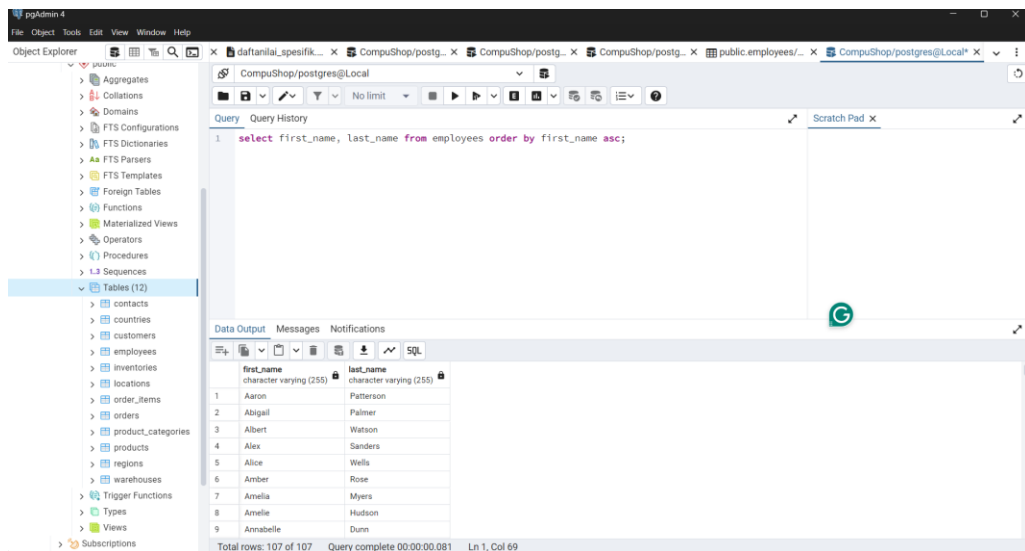
Memanggil nama kolom dengan sebutan nama lain



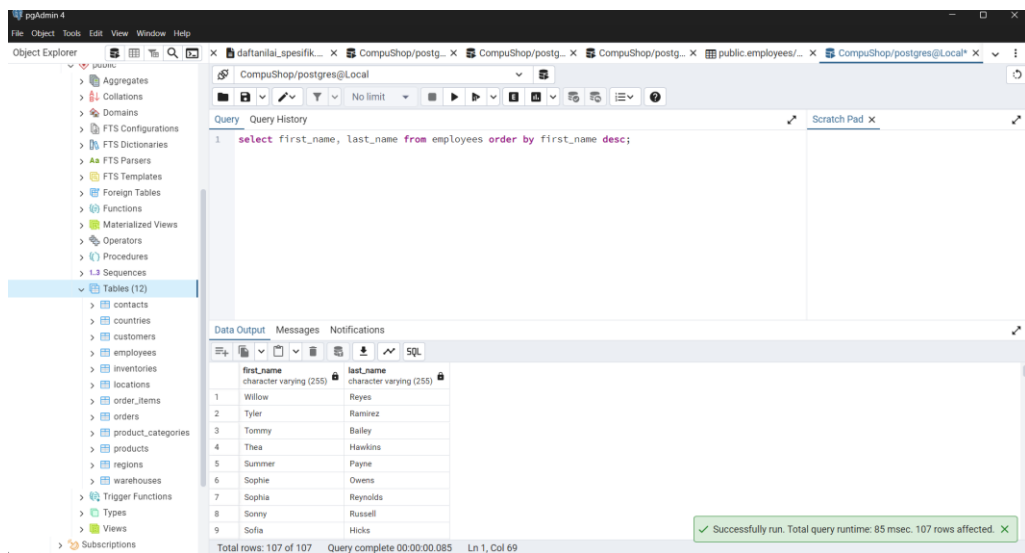
5. Order By

Mengurutkan data

- Ascending



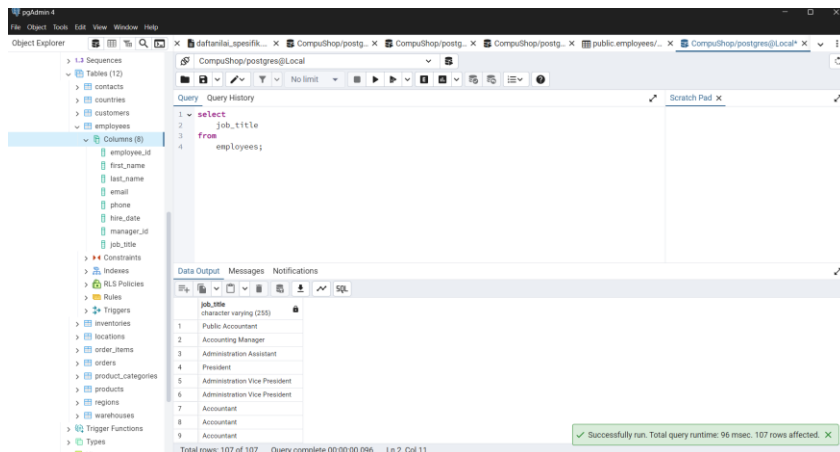
- Descending



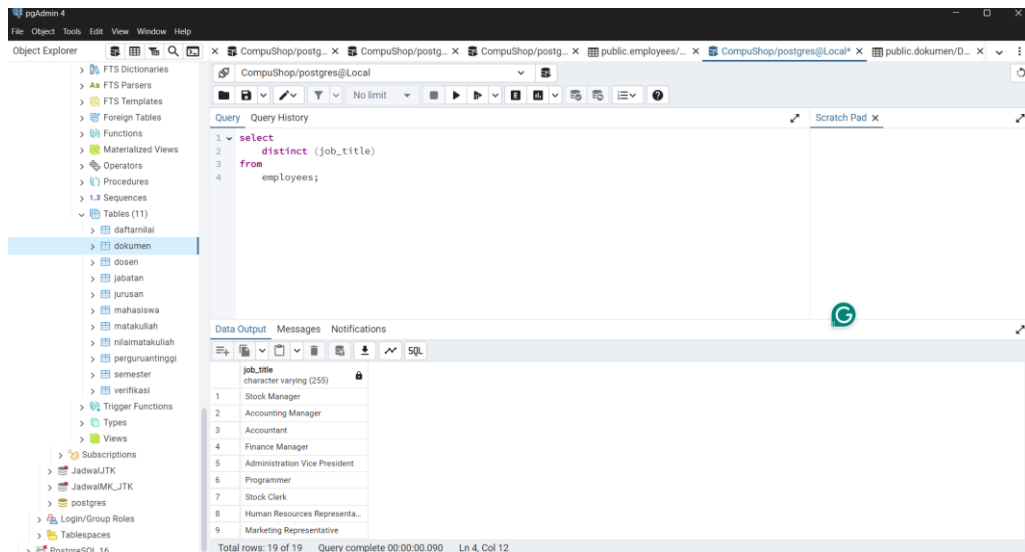
6. Select distinct

Mengambil data tanpa duplikasi

- Without distinct keyword

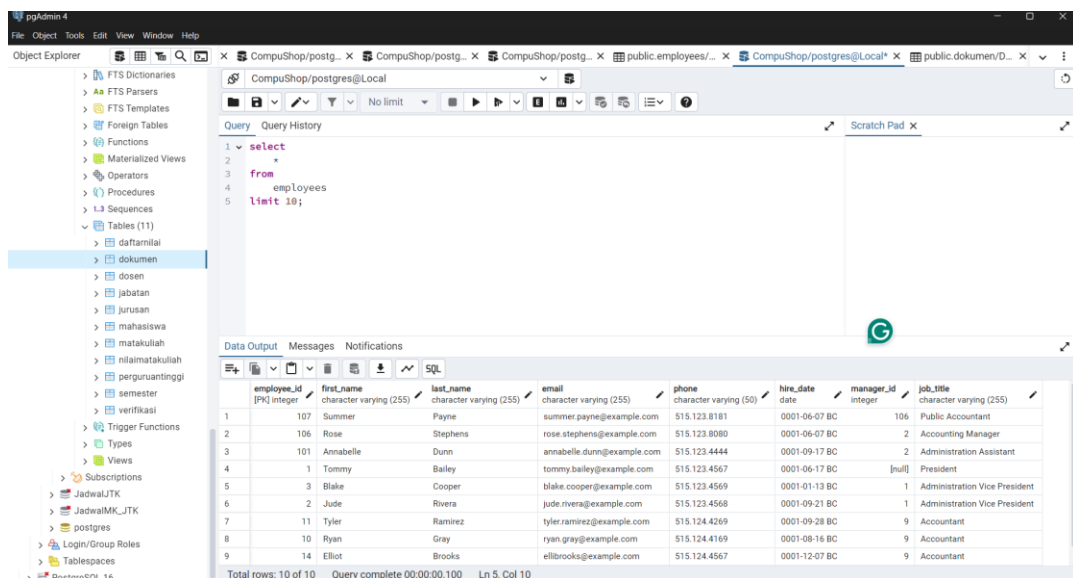


- With distinct keyword



7. Rownum atau Limit

Di PostgreSQL, tidak ada command ROWNUM, tapi adanya LIMIT untuk membatasi berapa banyak data yang ditampilkan



8. Equal

- Equal

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

CompuShop/postgres@Local

Query

Query History

Scratch Pad

1 select * from employees

2 where hire_date = '0001-06-07 BC';

Data Output Messages Notifications

SQL

	employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
	PK integer	character varying (255)	character varying (255)	character varying (255)	integer varying (50)	date	integer	character varying (255)
1	107	Summer	Payne	summer.payne@example.com	515.123.8181	0001-06-07 BC	106	Public Accountant
2	106	Rose	Stephens	rose.stephens@example.com	515.123.8080	0001-06-07 BC	2	Accounting Manager
3	104	Harper	Spencer	harper.spencer@example.com	515.123.7777	0001-06-07 BC	2	Human Resources Representative
4	105	Gracie	Gardner	gracie.gardner@example.com	515.123.8888	0001-06-07 BC	2	Public Relations Representative

Total rows: 4 of 4 Query complete 00:00:00.187 Ln 2, Col 35

- Not Equal

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

File Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (12)

contacts

countries

customers

employees

Columns (8)

employee_id

first_name

last_name

email

phone

hire_date

manager_id

job_title

Constraints

Indexes

RLS Policies

Rules

Triggers

Inventories

locations

order_items

orders

product_categories

CompuShop/postgres@Local

Query Query History

Scratch Pad

1 select * from employees

2 where hire_date != '0001-06-07 BC';

Data Output Messages Notifications

SQL

	employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
	PK integer	character varying (255)	character varying (255)	character varying (255)	character varying (50)	date	integer	character varying (255)
1	101	Annabelle	Dunn	annabelle.dunn@example.com	515.123.4444	0001-09-17 BC	2	Administration Assistant
2	1	Tommy	Bailey	tommy.bailey@example.com	515.123.4567	0001-06-17 BC	[null]	President
3	3	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1	Administration Vice President
4	2	Jude	Rivera	jude.rivera@example.com	515.123.4568	0001-09-21 BC	1	Administration Vice President
5	11	Tyler	Ramirez	tyler.ramirez@example.com	515.124.4269	0001-09-28 BC	9	Accountant
6	10	Ryan	Gray	ryan.gray@example.com	515.124.4169	0001-08-16 BC	9	Accountant
7	14	Elliott	Brooks	elliott.brooks@example.com	515.124.4567	0001-12-07 BC	9	Accountant
8	12	Elliott	James	elliott.james@example.com	515.124.4369	0001-09-30 BC	9	Accountant
9	13	Albert	Watson	albert.watson@example.com	515.124.4469	0001-03-07 BC	9	Accountant

Total rows: 103 of 103 Query complete 00:00:00.098 Ln 2, Col 36

9. Greater Than and Less Than

- Greater than

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

CompuShop/postgres@Local

Query

Query History

Scratch Pad

Data Output Messages Notifications

SQL

	employee_id integer	first_name character varying (255)	last_name character varying (255)	email character varying (255)	phone character varying (50)	hire_date date	manager_id integer	job_title character varying (255)
1	107	Summer	Payne	summer.payne@example.com	515.123.8181	0001-06-07 BC	106	Public Accountant
2	106	Rose	Stephens	rose.stephens@example.com	515.123.8080	0001-06-07 BC	2	Accounting Manager
3	101	Annabelle	Dunn	annabelle.dunn@example.co..	515.123.4444	0001-09-17 BC	2	Administration Assistant
4	104	Harper	Spencer	harper.spencer@example.com	515.123.7777	0001-06-07 BC	2	Human Resources Representa...
5	102	Emma	Perkins	emma.perkins@example.com	515.123.5555	0001-02-17 BC	1	Marketing Manager
6	103	Amelle	Hudson	amelle.hudson@example.com	603.123.6666	0001-08-17 BC	102	Marketing Representative
7	105	Gracie	Gardner	gracie.gardner@example.com	515.123.8888	0001-06-07 BC	2	Public Relations Representative

Total rows: 7 of 7 Query complete 00:00:00.098 Ln 2, Col 25

- Less than

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
1 select * from employees
2 where employee_id < 100;
```

The Data Output pane shows the results of the query, which are 99 rows. The status bar indicates 'Total rows: 99 of 99' and 'Query complete 00:00:00.142 Ln 2, Col 25'.

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	Tommy	Bailey	tommy.bailey@example.com	515.123.4567	0001-06-17 BC		President
2	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1	Administration Vice President
3	Jude	Rivera	jude.rivera@example.com	515.123.4568	0001-09-21 BC	1	Administration Vice President
4	Tyler	Ramirez	tyler.ramirez@example.com	515.124.4269	0001-09-28 BC	9	Accountant
5	Ryan	Gray	ryan.gray@example.com	515.124.4169	0001-08-16 BC	9	Accountant
6	Elliot	Brooks	elliott.brooks@example.com	515.124.4567	0001-12-07 BC	9	Accountant
7	Elliott	James	elliott.james@example.com	515.124.4369	0001-09-30 BC	9	Accountant
8	Albert	Watson	albert.watson@example.com	515.124.4469	0001-03-07 BC	9	Accountant
9	Mohammad	Peterson	mohammad.peterson@example.com	515.124.4569	0001-06-17 BC	2	Finance Manager

10. Less or equal than (combining with order by)

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following query:

```
1 select * from employees
2 where employee_id <= 100
3 order by first_name asc;
```

The Data Output pane shows the results of the query, which are 100 rows. The status bar indicates 'Total rows: 100 of 100' and 'Query complete 00:00:00.116 Ln 3, Col 25'.

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	Aaron	Patterson	aaron.patterson@example.com	650.124.1334	0001-01-14 BC	21	Stock Clerk
2	Abigail	Palmer	abigail.palmer@example.com	650.505.4876	0001-12-19 BC	23	Shipping Clerk
3	Albert	Watson	albert.watson@example.com	515.124.4469	0001-03-07 BC	9	Accountant
4	Alex	Sanders	alex.sanders@example.com	515.127.4562	0001-05-18 BC	15	Purchasing Clerk
5	Alice	Wells	alice.wells@example.com	011.44.1346.329268	0001-01-24 BC	48	Sales Representative
6	Amber	Rose	amber.rose@example.com	650.507.9822	0001-05-23 BC	25	Shipping Clerk
7	Amelia	Myers	amelia.myers@example.com	650.121.8009	0001-10-17 BC	25	Stock Clerk
8	Austin	Flores	austin.flores@example.com	650.124.5234	0001-08-20 BC	22	Stock Clerk
9	Ava	Sullivan	ava.sullivan@example.com	011.44.1344.429268	0001-10-01 BC	1	Sales Manager

11. And

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where employee_id <= 100 and job_title = 'Stock Clerk';

```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	33	Reggie	Simmons	reggie.simmons@example.com	650.124.8234	0001-04-10 BC	22 Stock Clerk
2	44	Emily	Hamilton	emily.hamilton@example.com	650.121.2874	0001-03-15 BC	25 Stock Clerk
3	43	Olivia	Ford	olivia.ford@example.com	650.121.2994	0001-01-29 BC	25 Stock Clerk
4	42	Amelia	Myers	amelia.myers@example.com	650.121.8009	0001-10-17 BC	25 Stock Clerk
5	41	Connor	Hayes	connor.hayes@example.com	650.121.1834	0001-04-06 BC	24 Stock Clerk
6	26	Leon	Powell	leon.powell@example.com	650.124.1214	0001-07-16 BC	21 Stock Clerk
7	27	Kai	Long	kai.long@example.com	650.124.1224	0001-09-28 BC	21 Stock Clerk
8	28	Aaron	Patterson	aaron.patterson@example.com	650.124.1334	0001-01-14 BC	21 Stock Clerk
9	29	Roman	Hughes	roman.hughes@example.com	650.124.1434	0001-03-08 BC	21 Stock Clerk

Total rows: 20 of 20 Query complete 00:00:00.105 Ln 2, Col 56

12. Or

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where employee_id <= 50 or job_title = 'Stock Clerk';

```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
22	48	Jessica	Woods	jessica.woods@example.com	011.44.1344.429278	0001-03-10 BC	1 Sales Manager
23	47	Ella	Wallace	ella.wallace@example.com	011.44.1344.467268	0001-01-05 BC	1 Sales Manager
24	46	Ava	Sullivan	ava.sullivan@example.com	011.44.1344.429268	0001-10-01 BC	1 Sales Manager
25	50	Mia	West	mia.west@example.com	011.44.1344.429018	0001-01-29 BC	1 Sales Manager
26	33	Reggie	Simmons	reggie.simmons@example.com	650.124.8234	0001-04-10 BC	22 Stock Clerk
27	44	Emily	Hamilton	emily.hamilton@example.com	650.121.2874	0001-03-15 BC	25 Stock Clerk
28	43	Olivia	Ford	olivia.ford@example.com	650.121.2994	0001-01-29 BC	25 Stock Clerk
29	42	Amelia	Myers	amelia.myers@example.com	650.121.8009	0001-10-17 BC	25 Stock Clerk
30	41	Connor	Hayes	connor.hayes@example.com	650.121.1834	0001-04-06 BC	24 Stock Clerk

Total rows: 50 of 50 Query complete 00:00:00.222 Ln 2, Col 54

13. And and Or

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where (employee_id <= 50 or job_title = 'Stock Clerk') and manager_id < 5;

```

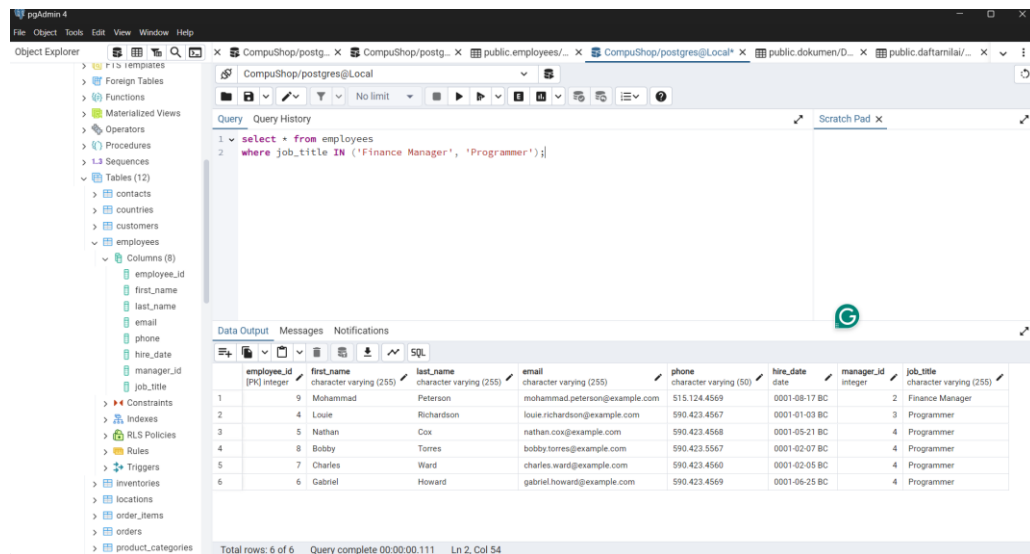
Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	3	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1 Administration Vice President
2	2	Jude	Rivera	jude.rivera@example.com	515.123.4568	0001-09-21 BC	1 Administration Vice President
3	9	Mohammad	Peterson	mohammad.peterson@example.com	515.124.4569	0001-08-17 BC	2 Finance Manager
4	4	Louie	Richardson	louie.richardson@example.com	590.423.4567	0001-01-03 BC	3 Programmer
5	5	Nathan	Cox	nathan.cox@example.com	590.423.4568	0001-05-21 BC	4 Programmer
6	8	Bobby	Torres	bobby.torres@example.com	590.423.5567	0001-02-07 BC	4 Programmer
7	7	Charles	Ward	charles.ward@example.com	590.423.4560	0001-02-05 BC	4 Programmer
8	6	Gabriel	Howard	gabriel.howard@example.com	590.423.4569	0001-06-25 BC	4 Programmer
9	15	Rory	Kelly	rory.kelly@example.com	515.127.4561	0001-12-07 BC	1 Purchasing Manager

Total rows: 19 of 19 Query complete 00:00:00.103 Ln 2, Col 75

14. In and Not In

- In



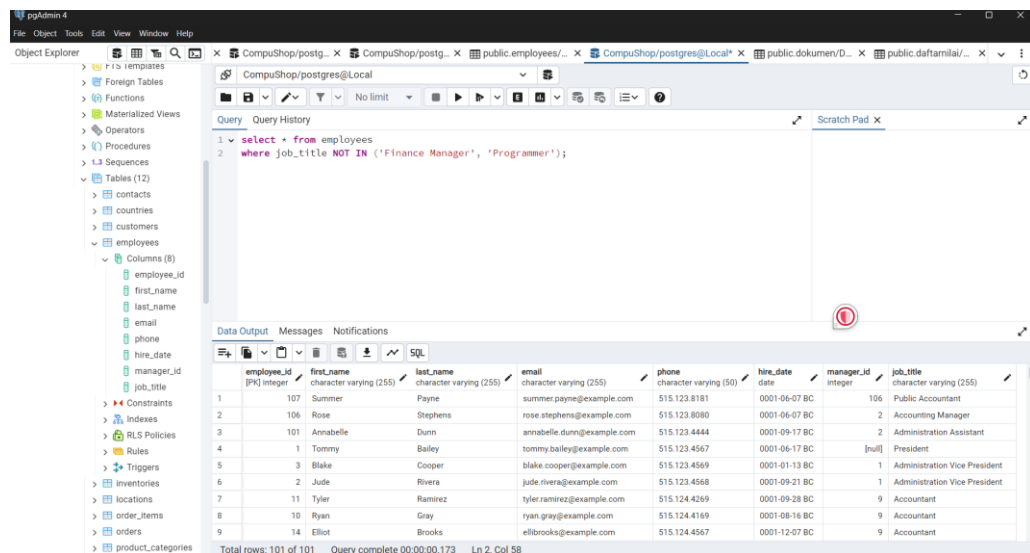
The screenshot shows the pgAdmin 4 interface with a query window open. The query is:

```
1 select * from employees
2 where job_title IN ('Finance Manager', 'Programmer');
```

The Data Output tab shows the results of the query, which are 6 rows. The status bar indicates "Total rows: 6 of 6" and "Query complete 00:00:00.111".

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
9	Mohammad	Peterson	mohammad.peterson@example.com	515.124.4569	0001-08-17 BC	2	Finance Manager
4	Louie	Richardson	louie.richardson@example.com	590.423.4567	0001-01-03 BC	3	Programmer
5	Nathan	Cox	nathan.cox@example.com	590.423.4568	0001-05-21 BC	4	Programmer
8	Bobby	Torres	bobby.torres@example.com	590.423.5567	0001-02-07 BC	4	Programmer
7	Charles	Ward	charles.ward@example.com	590.423.4560	0001-02-05 BC	4	Programmer
6	Gabriel	Howard	gabriel.howard@example.com	590.423.4569	0001-06-25 BC	4	Programmer

- Not In



The screenshot shows the pgAdmin 4 interface with a query window open. The query is:

```
1 select * from employees
2 where job_title NOT IN ('Finance Manager', 'Programmer');
```

The Data Output tab shows the results of the query, which are 101 rows. The status bar indicates "Total rows: 101 of 101" and "Query complete 00:00:00.173".

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
107	Summer	Payne	summer.payne@example.com	515.123.8181	0001-06-07 BC	106	Public Accountant
106	Rose	Stephens	rose.stephens@example.com	515.123.6080	0001-06-07 BC	2	Accounting Manager
101	Annabelle	Dunn	annabelle.dunn@example.com	515.123.4444	0001-09-17 BC	2	Administration Assistant
1	Tommy	Bailey	tommy.bailey@example.com	515.123.4567	0001-06-17 BC	[null]	President
3	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1	Administration Vice President
2	Jude	Rivera	jude.rivera@example.com	515.123.4568	0001-09-21 BC	1	Administration Vice President
11	Tyler	Ramirez	tyler.ramirez@example.com	515.124.4269	0001-09-28 BC	9	Accountant
10	Ryan	Gray	ryan.gray@example.com	515.124.4169	0001-08-16 BC	9	Accountant
14	Elliot	Brooks	elliott.brooks@example.com	515.124.4567	0001-12-07 BC	9	Accountant

15. Between and Not Between

- Between

pgAdmin 4

Object Explorer

- CompuShop/postgres@Local
 - Tables (12)
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title

Query

```
1 select * from employees
2 where employee_id between 100 and 105;
```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	101	Annabelle	Dunn	annabelle.dunn@example.co...	515.123.4444	0001-09-17 BC	2
2	104	Harper	Spencer	harper.spencer@example.com	515.123.7777	0001-06-07 BC	2
3	102	Emma	Perkins	emma.perkins@example.com	515.123.5555	0001-02-17 BC	1
4	103	Amelle	Hudson	amelle.hudson@example.com	603.123.6666	0001-08-17 BC	102
5	105	Gracie	Gardner	gracie.gardner@example.com	515.123.8888	0001-06-07 BC	2
6	100	Thea	Hawkins	thea.hawkins@example.com	650.507.9844	0001-01-13 BC	25

Successfully run. Total query runtime: 97 msec. 6 rows affected.

- Not Between

pgAdmin 4

Object Explorer

- CompuShop/postgres@Local
 - Tables (12)
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title

Query

```
1 select * from employees
2 where employee_id not between 100 and 105;
```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	107	Summer	Payne	summer.payne@example.com	515.123.8181	0001-06-07 BC	106
2	106	Rose	Stephens	rose.stephens@example.com	515.123.8080	0001-06-07 BC	2
3	1	Tommy	Bailey	tommy.bailey@example.com	515.123.4567	0001-06-17 BC	[null]
4	3	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1
5	2	Jude	Rivers	jude.rivers@example.com	515.123.4568	0001-09-21 BC	1
6	11	Tyler	Ramirez	tyler.ramirez@example.com	515.124.4269	0001-09-28 BC	9
7	10	Ryan	Gray	ryan.gray@example.com	515.124.4169	0001-08-16 BC	9
8	14	Elliott	Brooks	elliott.brooks@example.com	515.124.4567	0001-12-07 BC	8
9	12	Elliott	James	elliott.james@example.com	515.124.4567	0001-12-07 BC	8

Successfully run. Total query runtime: 95 msec. 101 rows affected.

16. Is Null and Is Not Null

- Is Null

pgAdmin 4

Object Explorer

- CompuShop/postgres@Local
 - Tables (12)
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title

Query

```
1 select * from employees
2 where manager_id IS NULL;
```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
1	1	Tommy	Bailey	tommy.bailey@example.com	515.123.4567	0001-06-17 BC	[null]

Successfully run. Total query runtime: 00:00:00.247 Ln 2, Col 26

- Is Not Null

The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL query:

```
1 select * from employees
2 where manager_id IS NOT NULL;
```

The 'Data Output' tab shows the results of the query. The status bar at the bottom indicates: 'Total rows: 106 of 106 Query complete 00:00:00.303 Ln 2, Col 30'. A green notification box at the bottom right states: 'Successfully run. Total query runtime: 299 msec. 106 rows affected.'

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
107	Summer	Payne	summer.payne@example.com	515.123.8181	0001-06-07 BC	106	Public Accountant
106	Rose	Stephens	rose.stephens@example.com	515.123.8080	0001-06-07 BC	2	Accounting Manager
101	Annabelle	Dunn	annabelle.dunn@example.com	515.123.4444	0001-09-17 BC	2	Administration Assistant
3	Blake	Cooper	blake.cooper@example.com	515.123.4569	0001-01-13 BC	1	Administration Vice President
2	Jude	Rivera	jude.rivera@example.com	515.123.4568	0001-09-21 BC	1	Administration Vice President
11	Tyler	Ramirez	tyler.ramirez@example.com	515.124.4269	0001-09-28 BC	9	Accountant
10	Ryan	Gray	ryan.gray@example.com	515.124.4169	0001-08-16 BC	9	Accountant
14	Elliott	Brooks	elliott.brooks@example.com	515.124.4567	0001-12-07 BC	9	Accountant
12	Elliott	James	elliott.james@example.com	515.124.4567	0001-12-07 BC	9	Accountant

17. Like

- Start with

The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL query:

```
1 select * from employees
2 where first_name LIKE 'N%';
```

The 'Data Output' tab shows the results of the query. The status bar at the bottom indicates: 'Total rows: 1 of 1 Query complete 00:00:00.106 Ln 2, Col 29'. A green notification box at the bottom right states: 'Successfully run. Total query runtime: 299 msec. 106 rows affected.'

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
5	Nathan	Cox	nathan.cox@example.com	590.423.4568	0001-05-21 BC	4	Programmer

- End with

pgAdmin 4

Object Explorer

- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where first_name LIKE 'lna%';

```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
67	Sienna	Simpson	sienna.simpson@example.com	011.44.1346.629268	0001-03-24 BC	48	Sales Representative

Total rows: 1 of 1 Query complete 00:00:00.108 Ln 2, Col 29

- Have

pgAdmin 4

Object Explorer

- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where first_name LIKE 'lna%';

```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
101	Annabelle	Dunn	annabelle.dunn@example.com	515.123.4444	0001-09-17 BC	2	Administration Assistant
67	Sienna	Simpson	sienna.simpson@example.com	011.44.1346.629268	0001-03-24 BC	48	Sales Representative
96	Hannah	Knight	hannah.knight@example.com	650.501.4876	0001-03-17 BC	24	Shipping Clerk

Successfully run. Total query runtime: 86 msec. 3 rows affected.

Total rows: 3 of 3 Query complete 00:00:00.086 Ln 2, Col 30

- Lowercase and uppercase

pgAdmin 4

Object Explorer

- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 select * from employees
2 where LOWER(first_name) LIKE 'lna%';

```

Data Output

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
101	Annabelle	Dunn	annabelle.dunn@example.com	515.123.4444	0001-09-17 BC	2	Administration Assistant
5	Nathan	Cox	nathan.cox@example.com	590.423.4568	0001-05-21 BC	4	Programmer
67	Sienna	Simpson	sienna.simpson@example.com	011.44.1346.629268	0001-03-24 BC	48	Sales Representative
96	Hannah	Knight	hannah.knight@example.com	650.501.4876	0001-03-17 BC	24	Shipping Clerk

Total rows: 4 of 4 Query complete 00:00:00.086 Ln 2, Col 37

18. Concat

Menggabungkan dua nilai string pada tabel

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following code:

```
1 SELECT
2     first_name,
3     last_name,
4     (first_name || ' ' || last_name) AS full_name
5 FROM
6     employees;
```

The Data Output pane displays the results of the query:

first_name	last_name	full_name
1 Summer	Payne	Summer Payne
2 Rose	Stephens	Rose Stephens
3 Annabelle	Dunn	Annabelle Dunn
4 Tommy	Bailey	Tommy Bailey
5 Blake	Cooper	Blake Cooper
6 Jude	Rivera	Jude Rivera
7 Tyler	Ramirez	Tyler Ramirez
8 Ryan	Gray	Ryan Gray
9 Elliot	Brooks	Elliot Brooks

Total rows: 107 of 107 Query complete 00:00:00.095 Ln 6, Col 12

19. Upper and Lower

The screenshot shows the pgAdmin 4 interface. The SQL query editor contains the following code:

```
1 SELECT
2     first_name,
3     upper(first_name) as upper_first_name,
4     lower(first_name) as lower_first_name
5 FROM
6     employees;
```

The Data Output pane displays the results of the query:

first_name	upper_first_name	lower_first_name
1 Summer	SUMMER	summer
2 Rose	ROSE	rose
3 Annabelle	ANNABELLE	annabelle
4 Tommy	TOMMY	tommy
5 Blake	BLAKE	blake
6 Jude	JUDE	jude
7 Tyler	TYLER	tyler
8 Ryan	RYAN	ryan
9 Elliot	ELLIOT	elliott

Total rows: 107 of 107 Query complete 00:00:00.090 Ln 4, Col 39

20. Length

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 SELECT
2   first_name,
3   length(first_name) as panjang_first_name
4 FROM
5   employees;

```

Data Output

first_name	panjang_first_name
1 Summer	6
2 Rose	4
3 Annabelle	9
4 Tommy	5
5 Blake	5
6 Jude	4
7 Tyler	5
8 Ryan	4
9 Elliot	6

Total rows: 107 of 107 Query complete 00:00:00.075 Ln 3, Col 42

Successfully run. Total query runtime: 75 msec. 107 rows affected.

21. Initcap (string value)

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 SELECT
2   first_name,
3   email,
4   INITCAP (email) as initcap_email
5 FROM
6   employees;

```

Data Output

first_name	email	initcap_email
1 Summer	summer.payne@example.com	Summer.Payne@Example.Com
2 Rose	rose.stephens@example.com	Rose.Stephens@Example.Com
3 Annabelle	annabelle.dunn@example.com	Annabelle.Dunn@Example.Com
4 Tommy	tommy.bailey@example.com	Tommy.Bailey@Example.Com
5 Blake	blake.cooper@example.com	Blake.Cooper@Example.Com
6 Jude	jude.rivera@example.com	Jude.Rivera@Example.Com
7 Tyler	tyler.ramirez@example.com	Tyler.Ramirez@Example.Com
8 Ryan	ryan.gray@example.com	Ryan.Gray@Example.Com
9 Elliot	ellibrooks@example.com	Ellibrooks@Example.Com

Total rows: 107 of 107 Query complete 00:00:00.099 Ln 6, Col 12

22. Substring

pgAdmin 4

Object Explorer

- Templates
- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (12)
 - contacts
 - countries
 - customers
 - employees
 - Columns (8)
 - employee_id
 - first_name
 - last_name
 - email
 - phone
 - hire_date
 - manager_id
 - job_title
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - inventories
 - locations
 - order_items
 - orders
 - product_categories

Query

```

1 SELECT
2   first_name,
3   hire_date,
4   substring(to_char(hire_date, 'YYYY-MM-DD BC'), 1, 4) AS year
5 FROM
6   employees;

```

Data Output

first_name	hire_date	year
1 Summer	0001-06-07 BC	0001
2 Rose	0001-06-07 BC	0001
3 Annabelle	0001-09-17 BC	0001
4 Tommy	0001-06-17 BC	0001
5 Blake	0001-01-13 BC	0001
6 Jude	0001-09-21 BC	0001
7 Tyler	0001-09-28 BC	0001
8 Ryan	0001-08-16 BC	0001
9 Elliot	0001-12-07 BC	0001

Total rows: 107 of 107 Query complete 00:00:00.077 Ln 6, Col 12

23. TRIM

- LTRIM, RTRIM, TRIM

The screenshot shows the pgAdmin 4 interface with a SQL query editor. The query is as follows:

```
1 SELECT
2     'JTK' AS jtk,
3     LTRIM('JTK') AS ltrim_jtk,
4     RTRIM('JTK') AS rtrim_jtk,
5     TRIM('JTK') AS trim_jtk
6 FROM
7     employees;
```

The Data Output tab shows the results of the query:

	jtk	ltrim_jtk	rtrim_jtk	trim_jtk
1	JTK	JTK	JTK	JTK
2	JTK	JTK	JTK	JTK
3	JTK	JTK	JTK	JTK
4	JTK	JTK	JTK	JTK
5	JTK	JTK	JTK	JTK
6	JTK	JTK	JTK	JTK
7	JTK	JTK	JTK	JTK
8	JTK	JTK	JTK	JTK
9	JTK	JTK	JTK	JTK

Total rows: 107 of 107 Query complete 00:00:00.084 Ln 7, Col 12

- LTRIM (string value, trim_text) dan RTRIM (text)

The screenshot shows the pgAdmin 4 interface with a SQL query editor. The query is as follows:

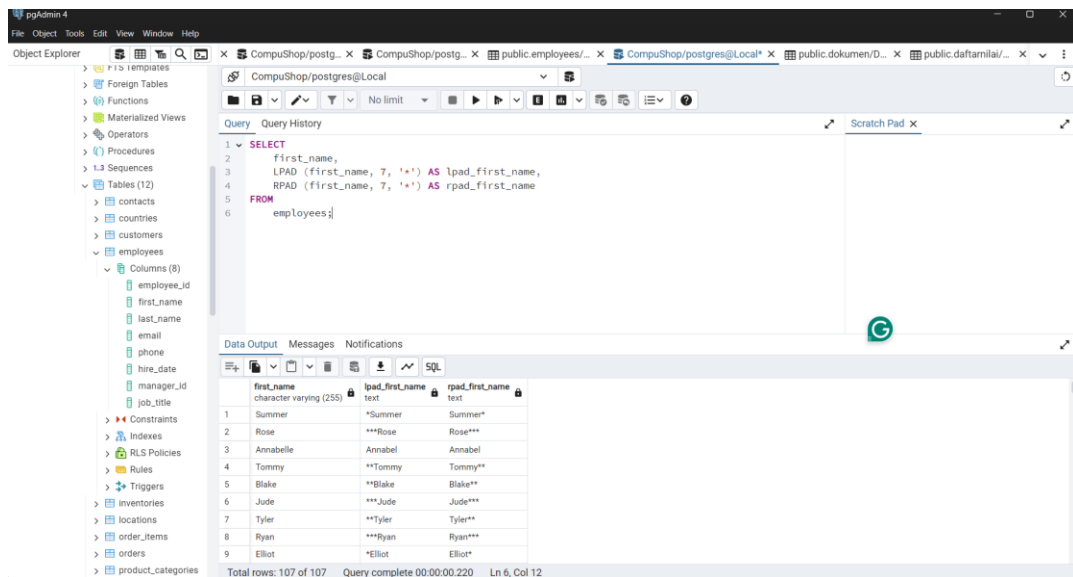
```
1 SELECT
2     job_title,
3     LTRIM(job_title, 'Pro') AS ltrim_job_title,
4     RTRIM(job_title, 'gamer') AS rtrim_job_title
5 FROM
6     employees
7 WHERE job_title = 'Programmer';
```

The Data Output tab shows the results of the query:

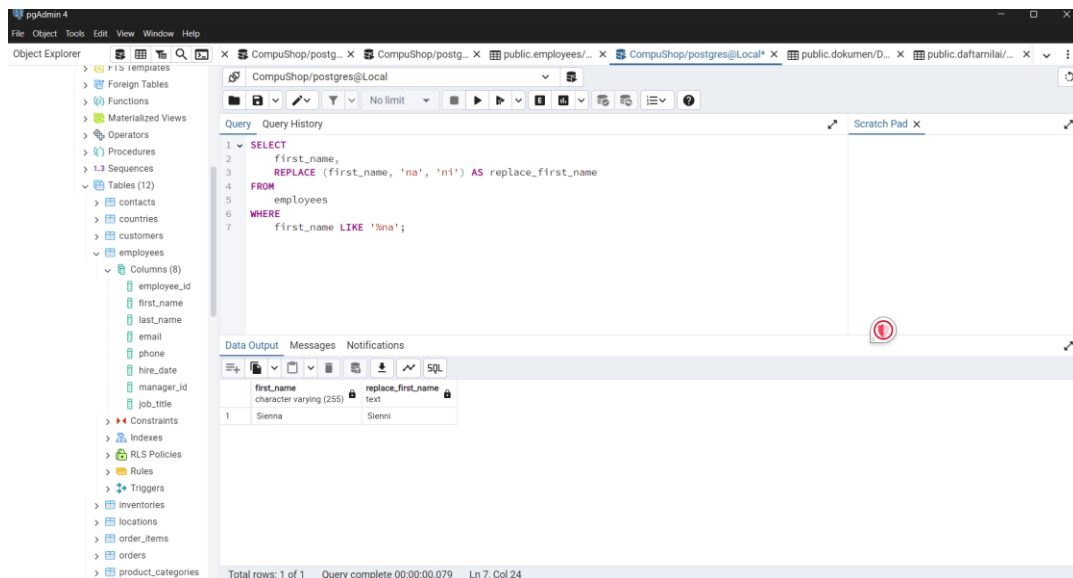
	job_title	ltrim_job_title	rtrim_job_title
1	Programmer	grammer	Pro
2	Programmer	grammer	Pro
3	Programmer	grammer	Pro
4	Programmer	grammer	Pro
5	Programmer	grammer	Pro

Total rows: 5 of 5 Query complete 00:00:00.091 Ln 7, Col 31

24. LPAD dan RPAD



25. Replace



26. Date Function

- ADD_MONTH, MONTHS_BETWEEN, ROUND, TRUNC

pgAdmin 4

Object Explorer

Query

```

1 SELECT
2   first_name,
3   date_trunc('day', hire_date) AS hire_date,
4   date_trunc('day', hire_date) + interval '2 month' AS add_month_hd,
5   EXTRACT(YEAR FROM age(date_trunc('day', hire_date), '2003-08-17')) * 12 + EXTRACT(MONTH FROM age(
6   date_trunc('month', hire_date) AS trunc_month_hd,
7   date_trunc('year', hire_date) AS trunc_year_hd,
8   date_trunc('month', hire_date) AS round_month_hd,
9   date_trunc('year', hire_date) AS round_year_hd
10  FROM
11  employees;

```

Data Output

name	hire_date	add_month_hd	trunc_month_hd	trunc_year_hd	round_month_hd	round_year_hd
1	0001-06-07 00:00:00+06:42:04 BC	0001-08-07 00:00:00+06:42:04 BC	-24038	0001-06-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-06-01 00:00:00+06:42:04 BC
2	0001-06-07 00:00:00+06:42:04 BC	0001-08-07 00:00:00+06:42:04 BC	-24038	0001-06-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-06-01 00:00:00+06:42:04 BC
3	0001-09-17 00:00:00+06:42:04 BC	0001-11-17 00:00:00+06:42:04 BC	-24035	0001-09-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-09-01 00:00:00+06:42:04 BC
4	0001-06-17 00:00:00+06:42:04 BC	0001-08-17 00:00:00+06:42:04 BC	-24038	0001-06-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-06-01 00:00:00+06:42:04 BC
5	0001-01-13 00:00:00+06:42:04 BC	0001-03-13 00:00:00+06:42:04 BC	-24043	0001-01-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC
6	0001-09-21 00:00:00+06:42:04 BC	0001-11-21 00:00:00+06:42:04 BC	-24034	0001-09-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-09-01 00:00:00+06:42:04 BC
7	0001-09-28 00:00:00+06:42:04 BC	0001-11-28 00:00:00+06:42:04 BC	-24034	0001-09-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-09-01 00:00:00+06:42:04 BC
8	0001-08-16 00:00:00+06:42:04 BC	0001-10-16 00:00:00+06:42:04 BC	-24036	0001-08-01 00:00:00+06:42:04 BC	0001-01-01 00:00:00+06:42:04 BC	0001-08-01 00:00:00+06:42:04 BC

Total rows: 107 of 107 Query complete 00:00:00.113 Ln 8, Col 51

- NEXT_DAY, LAST_DAY, SYSDATE, CURRENT_TIMESTAMP

pgAdmin 4

Object Explorer

Query

```

1 SELECT
2   first_name,
3   hire_date,
4   DATE_TRUNC('day', hire_date + INTERVAL '1 day') AS hd_next_day,
5   DATE_TRUNC('month', hire_date) AS hd_last_day,
6   CURRENT_DATE AS tanggal_skrng,
7   CURRENT_TIMESTAMP AS tanggal_dan_jam_sekarang
8  FROM
9  employees;

```

Data Output

first_name	hire_date	hd_next_day	hd_last_day	tanggal_skrng	tanggal_dan_jam_sekarang
1 Summer	0001-06-07 BC	0001-06-08 00:00:00 BC	0001-06-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
2 Rose	0001-06-07 BC	0001-06-08 00:00:00 BC	0001-06-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
3 Annabelle	0001-09-17 BC	0001-09-18 00:00:00 BC	0001-09-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
4 Tommy	0001-06-17 BC	0001-06-18 00:00:00 BC	0001-06-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
5 Blake	0001-01-13 BC	0001-01-14 00:00:00 BC	0001-01-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
6 Jude	0001-09-21 BC	0001-09-22 00:00:00 BC	0001-09-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
7 Tyler	0001-09-28 BC	0001-09-29 00:00:00 BC	0001-09-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
8 Ryan	0001-08-16 BC	0001-08-17 00:00:00 BC	0001-08-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07
9 Elliot	0001-12-07 BC	0001-12-08 00:00:00 BC	0001-12-01 00:00:00+06:42:04 BC	2024-09-17	2024-09-17 10:28:05.986776+07

Total rows: 107 of 107 Query complete 00:00:00.082 Ln 9, Col 12

- NEXT_DAY

pgAdmin 4

Object Explorer

Query

```

1 SELECT
2   CURRENT_DATE,
3   TO_CHAR(CURRENT_TIMESTAMP, 'DD-MM-YYYY HH24:MI:SS') AS tanggal_dan_jam_sekarang,
4   DATE_TRUNC('day', CURRENT_DATE + INTERVAL '1 day') AS rabu,
5   DATE_TRUNC('day', CURRENT_DATE + INTERVAL '3 days') AS jumat,
6   DATE_TRUNC('day', CURRENT_DATE + INTERVAL '6 days') AS senin
7  FROM
8  employees;

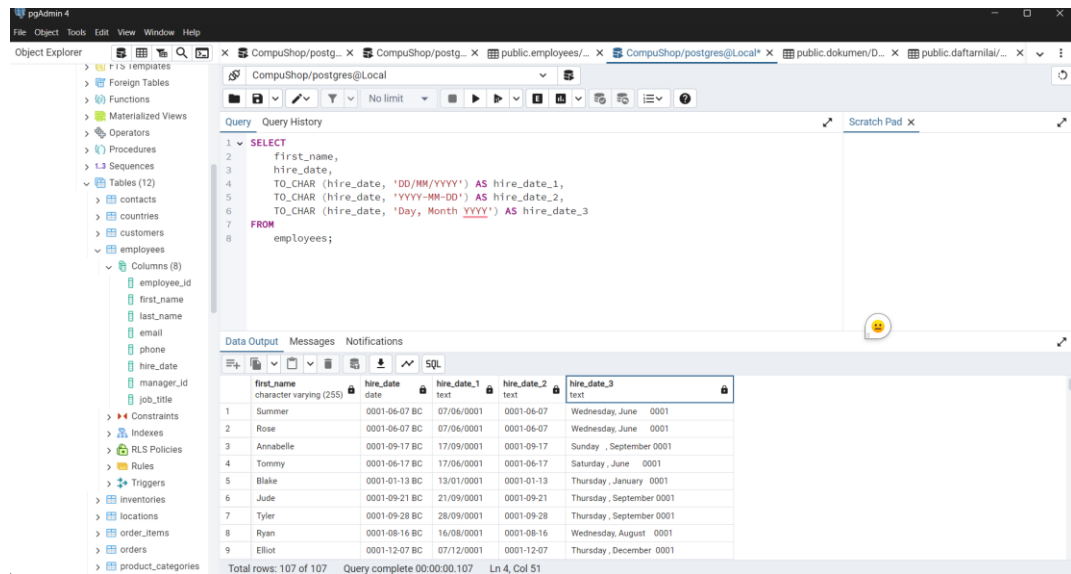
```

Data Output

current_date	tanggal_dan_jam_sekarang	rabu	jumat	senin
1 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
2 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
3 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
4 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
5 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
6 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
7 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
8 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00
9 2024-09-17	17-09-2024 10:43:09	2024-09-18 00:00:00	2024-09-20 00:00:00	2024-09-23 00:00:00

Total rows: 107 of 107 Query complete 00:00:00.085 Ln 6, Col 62

27. TO_CHAR



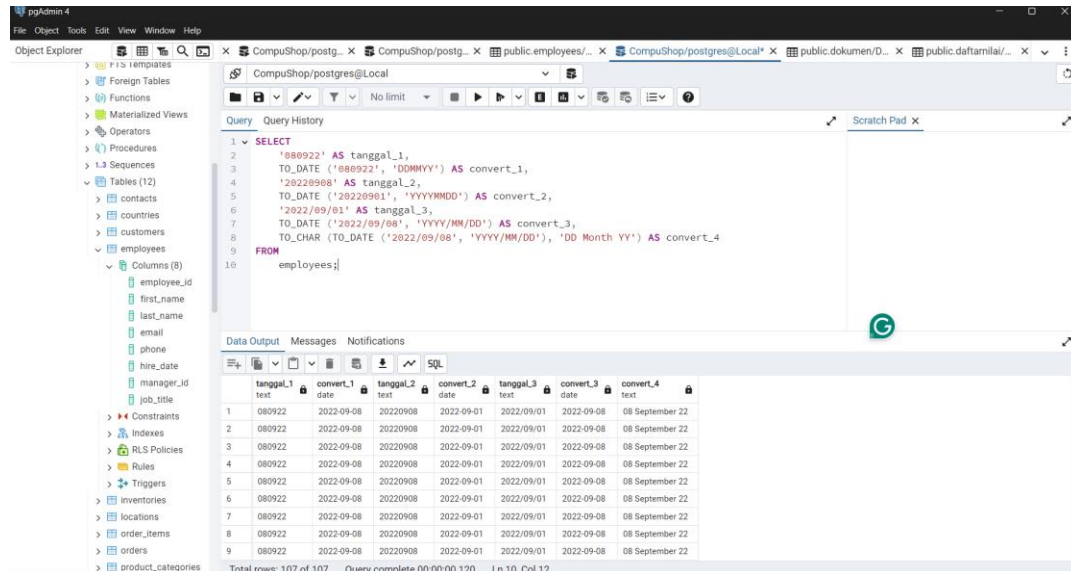
The screenshot shows the pgAdmin 4 interface with a SQL query executed against the 'employees' table. The query uses the TO_CHAR function to format the hire_date into different string representations. The results are displayed in a table with 5 columns: first_name, hire_date, hire_date_1, hire_date_2, and hire_date_3.

```
SELECT
  first_name,
  hire_date,
  TO_CHAR (hire_date, 'DD/MM/YYYY') AS hire_date_1,
  TO_CHAR (hire_date, 'YYYY-MM-DD') AS hire_date_2,
  TO_CHAR (hire_date, 'Day, Month YYYY') AS hire_date_3
FROM
  employees;
```

first_name	hire_date	hire_date_1	hire_date_2	hire_date_3
1 Summer	0001-06-07 BC	07/06/0001	0001-06-07	Wednesday, June 0001
2 Rose	0001-06-07 BC	07/06/0001	0001-06-07	Wednesday, June 0001
3 Annabelle	0001-09-17 BC	17/09/0001	0001-09-17	Sunday, September 0001
4 Tommy	0001-06-17 BC	17/06/0001	0001-06-17	Saturday, June 0001
5 Blake	0001-01-13 BC	13/01/0001	0001-01-13	Thursday, January 0001
6 Jude	0001-09-21 BC	21/09/0001	0001-09-21	Thursday, September 0001
7 Tyler	0001-09-28 BC	28/09/0001	0001-09-28	Thursday, September 0001
8 Ryan	0001-08-16 BC	16/08/0001	0001-08-16	Wednesday, August 0001
9 Elliot	0001-12-07 BC	07/12/0001	0001-12-07	Thursday, December 0001

Total rows: 107 of 107 Query complete 00:00:00.107 Ln 4, Col 51

28. TO_DATE



The screenshot shows the pgAdmin 4 interface with a SQL query that converts string dates to actual dates using TO_DATE and then formats them back using TO_CHAR. The results are displayed in a table with 8 columns: tanggal_1, convert_1, tanggal_2, convert_2, tanggal_3, convert_3, convert_4, and convert_5.

```
SELECT
  '080922' AS tanggal_1,
  TO_DATE ('080922', 'DDMMYY') AS convert_1,
  '20220908' AS tanggal_2,
  TO_DATE ('20220908', 'YYYYMMDD') AS convert_2,
  '2022/09/01' AS tanggal_3,
  TO_DATE ('2022/09/01', 'YYYY/MM/DD') AS convert_3,
  TO_CHAR (TO_DATE ('2022/09/01', 'YYYY/MM/DD'), 'DD Month YY') AS convert_4
FROM
  employees;
```

tanggal_1	convert_1	tanggal_2	convert_2	tanggal_3	convert_3	convert_4	convert_5
1 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
2 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
3 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
4 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
5 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
6 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
7 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
8 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	
9 080922	2022-09-08	20220908	2022-09-01	2022/09/01	2022-09-08	08 September 22	

Total rows: 107 of 107 Query complete 00:00:00.120 Ln 10, Col 12

29. NVL

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'orders' selected under the 'CompuShop/postgres@Local' database. The main pane shows a SQL query:

```

1 SELECT
2   salesman_id,
3   COALESCE(salesman_id, 999) AS salesman_id_2
4 FROM
5   orders;

```

The 'Data Output' pane shows the results of the query:

salesman_id	salesman_id_2
19	64
20	64
21	999
22	999
23	999
24	999
25	999
26	999
27	999

Total rows: 105 of 105. Query complete 00:00:00.082. Ln 5, Col 9.

30. Decode

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'products' selected under the 'CompuShop/postgres@Local' database. The main pane shows a SQL query:

```

1 SELECT
2   product_name,
3   list_price,
4   CASE
5     WHEN list_price > 1000 THEN 'High'
6     WHEN list_price > 500 THEN 'Medium'
7     ELSE 'Low'
8   END AS decode_price
9 FROM
10  products
11 ORDER BY list_price ASC;

```

The 'Data Output' pane shows the results of the query:

product_name	list_price	decode_price
Asus PRIME X299-DELLUXE	487.30	Low
ASRock E3C224D4M-16RE	499.99	Low
Gigabyte X299 AORUS Gaming 9	499.99	Low
Gigabyte GA-Z270X-Gaming 9	503.98	Medium
Asus Rampage V Edition 10	519.99	Medium
Supermicro H8D06-F	525.99	Medium
MSI X99A GODLIKE GAMING CARBON	549.99	Medium
Intel Core i7-5930K	554.99	Medium
Asus Z10PE-D8 WS	561.59	Medium

Total rows: 288 of 288. Query complete 00:00:00.094. Ln 9, Col 6.

31. CASE

32. Mathematical Operation

Query

```

1 SELECT
2   list_price,
3   (list_price-14000) AS mins_price,
4   ABS (list_price-14000) AS abs_price,
5   (list_price/3) AS divide_price,
6   ROUND (list_price/3, 2) AS round_price,
7   FLOOR (list_price/3) AS floor_price,
8   CEIL (list_price/3) AS ceil_price,
9   MOD (list_price, 3) AS mod_price,
10  POWER (list_price, 2) AS power_price,
11  SQRT (list_price) AS sqrt_price
12 FROM
13  products;

```

Data Output

	list_price numeric (9,2)	mins_price numeric	abs_price numeric	divide_price numeric	round_price numeric	floor_price numeric	ceil_price numeric	mod_price numeric	power_price numeric	sqrt_price numeric
1	3410.46	-10589.54	10589.54	1136.8200000000000000	1136.82	1136	1137	2.46	11631237.411600000	58.399143829340512
2	2774.98	-11225.02	11225.02	924.9933333333333333	924.99	924	925	2.98	7700514.000400000	52.678078932322504
3	2660.72	-11339.28	11339.28	886.9066666666666667	886.91	886	887	2.72	7079430.918400000	51.582167461245752
4	2554.99	-11445.01	11445.01	851.6633333333333333	851.66	851	852	1.99	6527973.900100000	50.54690806480127
5	2501.69	-11498.31	11498.31	833.8966666666666667	833.90	833	834	2.69	6258452.856100000	50.016897144864954
6	2431.95	-11568.05	11568.05	810.5000000000000000	810.65	810	811	1.95	5914380.802500000	49.314805079205170
7	2377.09	-11622.91	11622.91	792.3633333333333333	792.36	792	793	1.09	5650556.868100000	48.755409956229473
8	2269.99	-11730.01	11730.01	756.6633333333333333	756.66	756	757	1.99	5152854.600100000	47.644412054300765
9	2259.99	-11740.01	11740.01	753.3300000000000000	753.33	753	754	0.99	5107554.800100000	47.539352120112028

Total rows: 288 of 288 Query complete 00:00:00.103 Ln 13, Col 11

33. Aggregate Functions

- COUNT, MIN, MAX, SUM, AVG

Query

```

1 SELECT
2   COUNT(*) count_,
3   MIN (list_price) AS min_price,
4   MAX (list_price) AS max_price,
5   SUM (list_price) AS sum_price,
6   AVG (list_price) AS avg_price,
7   (SUM(list_price)/COUNT(*)) avg_price2
8 FROM
9  products;

```

Data Output

	count_ bigint	min_price numeric	max_price numeric	sum_price numeric	avg_price numeric	avg_price2 numeric
1	288	15.55	8867.99	260133.52	903.241388888888889	903.241388888888889

Total rows: 1 of 1 Query complete 00:00:00.103 Ln 9, Col 11

- GROUP BY

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences (1)
 - Tables (11)
 - daftamail
 - dokumen
 - dosen
 - jabatan
 - jurusan
 - mahasiswa
 - matakuliah
 - nilaimatakuliah
 - perguruan tinggi

CompuShop/postgres@Local

Query Query History

```
1 SELECT
2   category_id,
3   COUNT(*) count_,
4   MIN (list_price) AS min_price,
5   MAX (list_price) AS max_price,
6   SUM (list_price) AS sum_price,
7   AVG (list_price) AS avg_price,
8   (SUM(list_price)/COUNT(*)) avg_price2
9 FROM
10  products
11 GROUP BY
12  category_id;
```

Data Output Messages Notifications

category_id	count_	min_price	max_price	sum_price	avg_price	avg_price2
integer	bigint	numeric	numeric	numeric	numeric	numeric
1	70	554.99	3410.46	97087.65	1386.9664285714285714	1386.9664285714285714
2	108	15.55	8867.99	68603.38	635.2164814814814815	635.2164814814814815
3	60	279.99	948.99	24137.59	402.2931666666666667	402.2931666666666667
4	50	739.99	5499.99	70304.90	1406.0980000000000000	1406.0980000000000000

Total rows: 4 of 4 Query complete 00:00:00.117 Ln 12, Col 14

34. INSERT

35.